NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY 8875/01

Paper 1 Multiple Choice

28 September 2011

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **11** printed pages and **1** blank pages..

[Turn over

1 Which pair of organelles has internal membranes?

A Nucleus and lysosome

B Ribosomes and nucleus

C Chloroplast and lysosome

D Chloroplasts and mitochondria

2 An actively growing cell is supplied with radioactive amino acids. Which cell component would first show an increase in radioactivity?

A Nucleus

B Golgi body

C Mitochondrion

D Rough Endoplasmic reticulum

3 Which molecule maintains the fluidity of the cell surface membrane?

A Glycolipid

B Cholesterol

C Glycoprotein

D Phospholipid

4 All of the following are directly associated with the lipid components of the plasma membrane **except**

A Glycerol backbone

B Peripheral proteins

C Hydrocarbon chains

D Cholesterol molecules

5 Which process(s) can take place in a root hair cell when oxygen is **not** available?

A Endocytosis only

B Active transport only

C Diffusion and osmosis

D Active transport and osmosis

6 Which of the following is a feature of exocytosis but not endocytosis?

A Secretion

B Vesicle formation

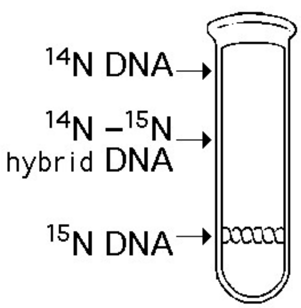
C Lipid bilayer fusion

D Lipid bilayer adhesion

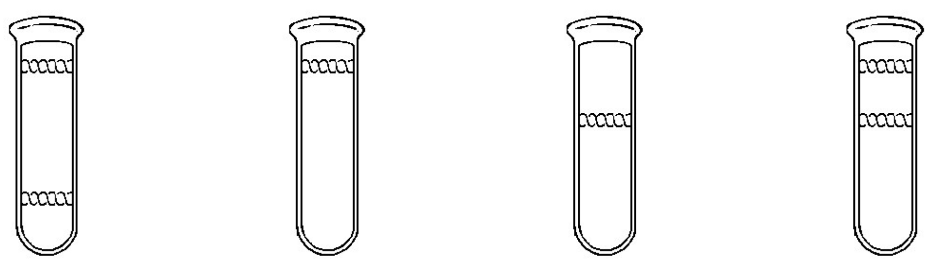
7 Which of the following shows the correct combination of bond(s) that need to be formed and the kind of reaction that is involved in order for the nucleotide to be added to the DNA chain?

	Bond(s) to be formed	Reaction(s) involved
A	Phosphoester	Dephosphorylation
B	Phosphodiester	Hydrolysis
C	Phosphodiester	Condensation and Dephosphorylation
D	Phosphoester and Hydrogen	Condensation

8 In the late 1950s, Meselson and Stahl grew bacteria in a medium containing "heavy" nitrogen (^{15}N) and then transferred them to a medium containing ^{14}N. Which of the following results would be expected after two rounds of DNA replication in the presence of ^{14}N?



Initial state



A **B** **C** **D**

9 What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversion of segments of chromosomes do occur.
C	Both affect DNA sequence.	Chromosomal aberrations may occur across chromosomes but gene mutations occur within a chromosome.
D	Both will not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

10 Which of the following structures would be found in an animal cell undergoing mitosis, but not in a plant cell undergoing mitosis?

A centromere

B centresome

C chromosome

D centriole

11 Which of the following is **<u>not true</u>** about cancerous cells?

A Ability to divide for a certain number of times.

B Ability to divide further when in contact with neighbouring cells.

C Inability to differentiate properly.

D Inability to exhibit anchorage dependence.

12 A cross between a tall plant and a dwarf plant, both with round leaves produced

121	round leaves, tall plants
124	round leaves, dwarf plants
42	oval leaves, tall plants
37	oval leaves, dwarf plants

key	
R	round leaf
r	oval leaf
T	tall
t	dwarf

What were the genotypes of the parents?

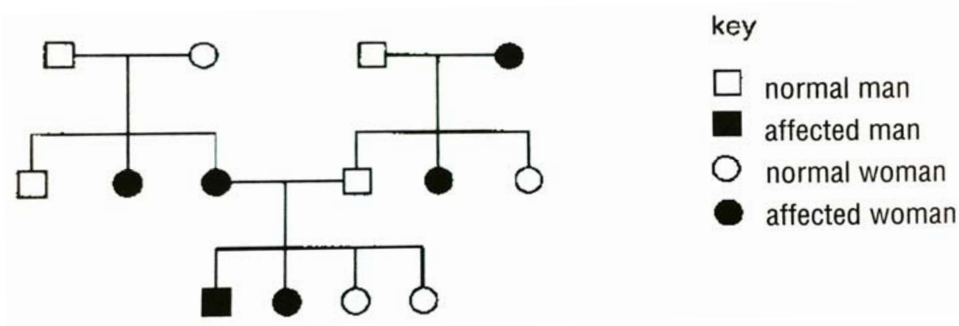
A RrTt x Rrtt

B RrTt x RRtt

C RrTT x Rrtt

D RrTT x RRtt

13 The family tree shows the inheritance of a skin condition.



What is the genetic basis of the skin condition?

A autosomal dominant

B sex-linked dominant

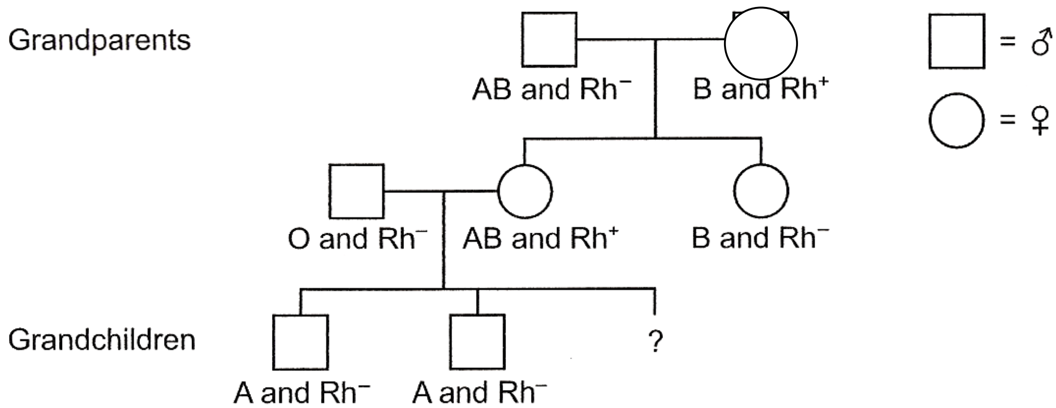
C autosomal recessive

D sex-linked recessive

14 A plant is heterozygous for a pair of alleles that are codominant. This plant is self-pollinated and the resulting seeds are germinated and allowed to grow. Which ratios are expected in the offspring?

	Ratio of phenotypes	Ratio of genotypes
A	1:2:1	1:2:1
B	1:2:1	3:1
C	3:1	1:2:1
D	3:1	3:1

15 The diagram shows a family tree.



What is the probability that the third grandchild will be a boy with blood group B and Rh positive blood?

A 0.0625

B 0.125

C 0.25

D 0.5

16 In lentils, the seed coat pattern is determined by a gene with 3 alleles. The phenotypes are marbled, spotted and clear. 4 crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

Cross	**Parents**	**Offspring phenotype and ratio**
1	marbled x marbled	3 marbled : 1 clear
2	spotted x clear	1 spotted : 1 clear
3	marbled x marbled	3 marbled : 1 spotted
4	clear x clear	all clear

From the data, it is possible to conclude that

A Spotted is recessive to clear.

B All of the clear offspring are heterozygous.

C Two-thirds of the marbled offspring in cross **3** are heterozygous.

D The marbled parents in cross **1** have the same genotype as the marbled parents in cross **3**.

17 Which one of the following phenotypic features of Man can be affected only by genotype?

A hair colour

B skin colour

C intelligence

D number of different blood group antigens

18 The following reaction sequence occurs in humans.

$$\text{Phenylalanine} \xrightarrow{\textbf{X}} \text{tyrosine} \xrightarrow{\textbf{Y}} \text{carbon dioxide and water}$$

Phenylketonuria (PKU) is a disease caused by an enzyme defiency in step **X**, and alkaptonuria (AKU) is due to an enzyme defiency in step **Y**. Both conditions are rare and caused by recessive alleles.

A person with PKU marries a person with AKU.

What phenotypes would be expected for their children?

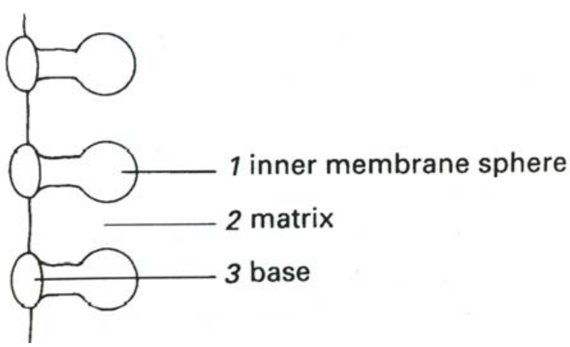
A all normal

B all AKU only

C all PKU only

D all both AKU and PKU

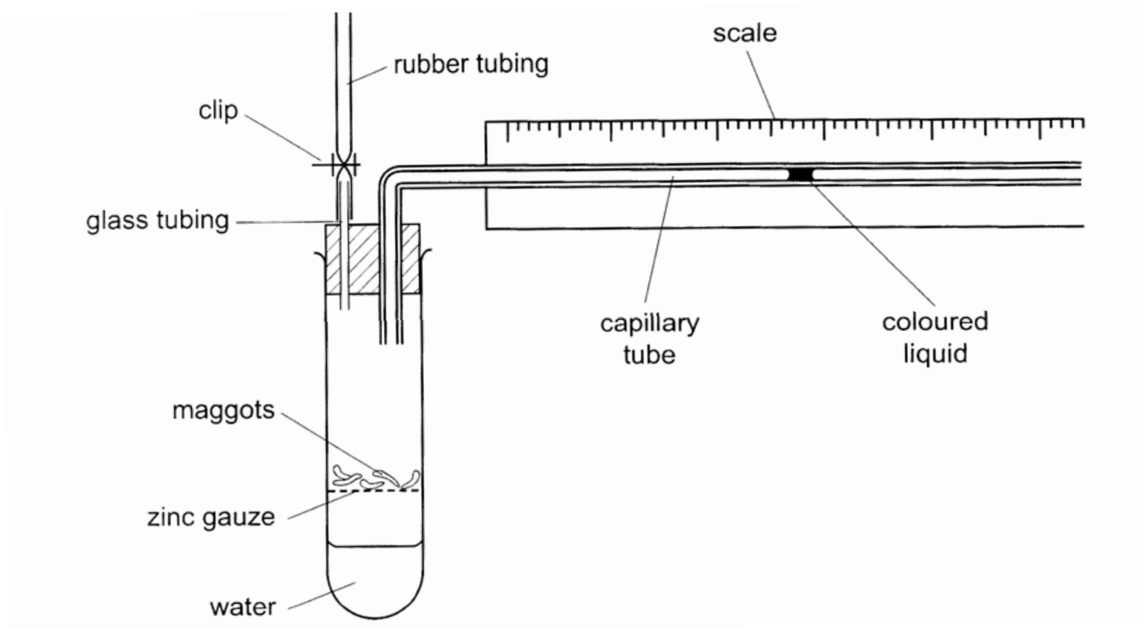
19 The diagram shows stalked particles on part of a crista membrane in a mitochondrion.



What process occurs in each of the numbered regions?

	1	2	3
A	ADP synthesis	electron transport	Krebs cycle
B	ADP synthesis	glycolysis	Krebs cycle
C	ATP synthesis	Krebs cycle	electron transport
D	ATP synthesis	Krebs cycle	glycolysis

20 The diagram shows a respirometer that is being used to investigate respiration in maggots.



The conditions are kept constant and the clip is closed. What happens to the coloured liquid when the maggots respire at an RQ of 1?

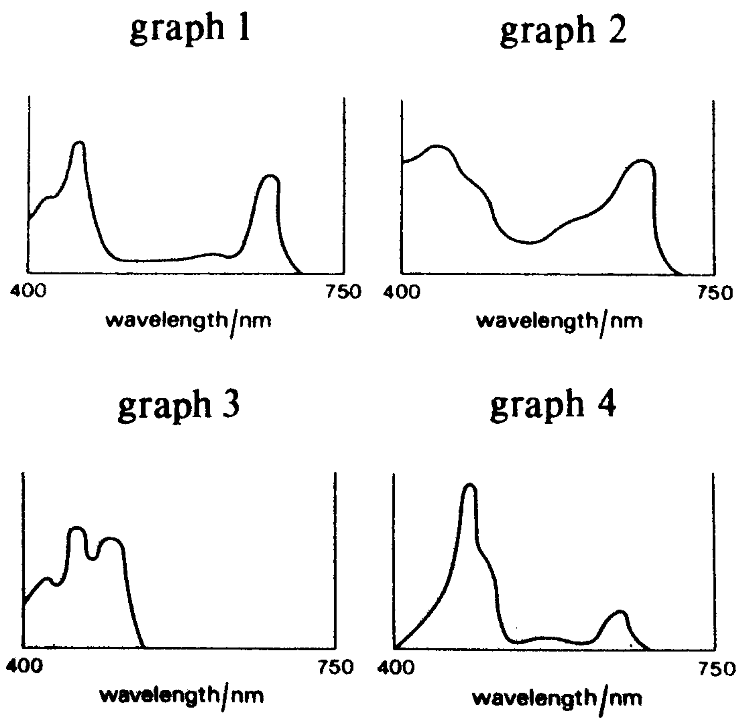
A It moves to the right continuously.

B It moves to the left continuously.

C It moves to the right and stops.

D It remains stationary.

21 Three of the four graphs below show the absorption spectra of photosynthetic pigments. One of the four graphs shows the action spectrum of photosynthesis for a plant containing those pigments.



Which one of the following series identifies the four graphs?

	Absorption spectra			***Action spectrum***
	chlorophyll a	***chlorophyll b***	***carotenoids***	
A	1	4	3	2
B	2	1	3	4
C	2	4	3	1
D	3	2	4	1

22 A group of small fish live in a lake with a uniformly light-brown sandy bottom. Most of the fish are light brown, but about 10% are mottled. The fish species is often prey for large birds that live on the shore. A construction company dumps a load of gravel in the bottom of the lake, giving it a mottled appearance.

Which of these statements presents the most accurate prediction of what will happen to this fish population?

A As the mottled fish are eaten, more will be produced to fill the gap.

B In two generations, all the fish will be mottled.

C The proportion of mottled fish will increase over time.

D The ratio of 9 light brown: 1 mottled fish will not change.

23 Which of the following pairs is **least** likely to represent homology?

A The haemoglobin of a baboon and that of a gorilla

B The mitochondria of a plant and those of an animal

C The wings of a bird and those of an insect

D The wings of a bat and the arms of a human

24 A piece of DNA is introduced into the ampicillin resistant gene of a plasmid. This plasmid also has a tetracycline resistant gene. Bacteria transformed with this plasmid will be

A resistant to both antibiotics

B sensitive to both antibiotics

C resistant to tetracycline only

D sensitive to tetracycline only

25 A student performed the following steps in a genetic engineering experiment to select for the recombinant bacterial colonies.

i. Cleaved DNA separated using gel electrophoresis.

ii. Isolation of gene of interest from genomic or cDNA library.

iii. Gene amplification via PCR.

iv. Insertion of genes into plasmids.

v. Transformation using heat shock.

vi. Selection via antibiotic resistance or blue-white screening.

Which is the correct sequence?

A ii → iii → i → iv → v → vi

B ii → iii → i → v → vi → iv

C iv → v → i → ii → vi → iii

D v → iv → iii → ii → i → vi

26 Which of the following characteristics is undesirable in cloning vectors used in genetic engineering?

A vulnerable at several sites to a restriction enzyme

B control their own replication

C high copy number

D small in size

27 Which of these processes could increase crop yield to alleviate food shortage problem?

A inserting genes for pest resistance into cotton

B inserting genes for human milk protein into cows

C inserting genes for vitamin A production into rice

D inserting genes for herbicide resistance into corn

28 Which of the following is **NOT** a goal of the Human Genome Project?

A Identify all alleles present in human beings.

B Map the genetic differences between individuals.

C Map and sequence all non-coding sequences in human beings.

D Develop computer software to analyse genetic information which had been sequenced.

29 Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to produce triploid salmon which are infertile.

Reproductive organ tissue from diploid trout was transplanted into the newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

A young trout

B triploid salmon

C salmon providing developing eggs

D trout providing reproductive organ tissue

30 Embryonic stem (ES) cells are an attractive source of material for therapeutic cloning because

A they can be induced to become any cell in the body.

B ES cells will not work as a source of tissue for cloning.

C ES cells are not targets for the host immune response, so tissue rejection is not an issue.

D there are no sources of stem cells to use for therapeutic cloning, so ES cells are the only solution.